

# **ENV 4120**

## **Air Pollution and Hazardous Waste Control**

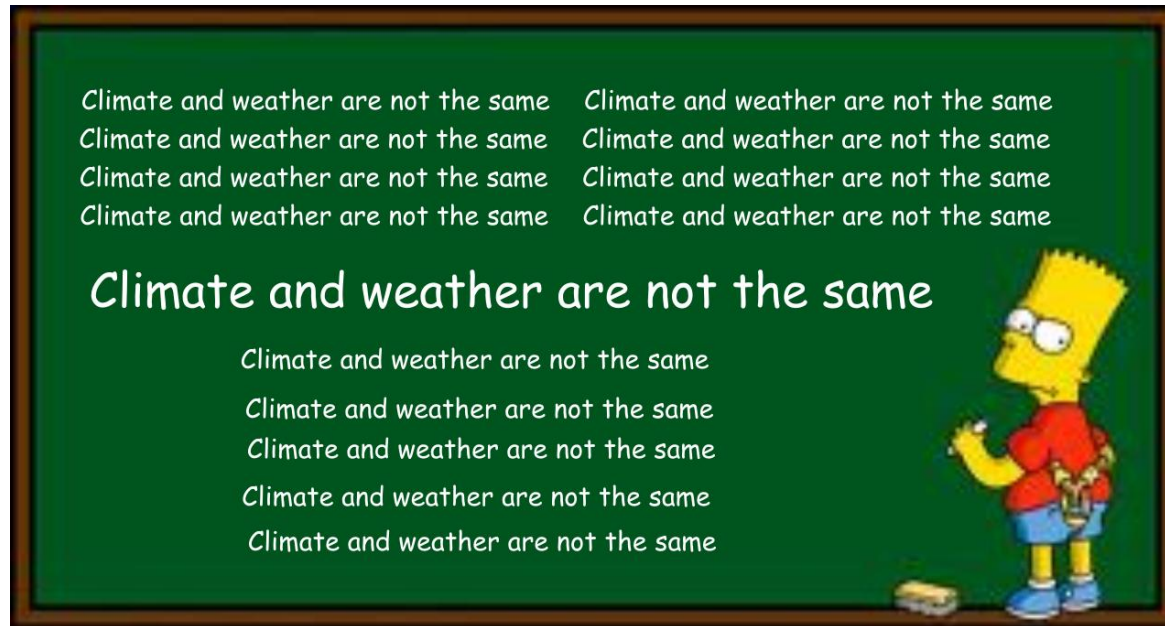
# Today's Topics

- Basic science related to climate change
  - Earth's energy balance
  - Milankovitch Cycles
  - Natural green house effect
  - Radiative forcing
  - Climate feedbacks
  - Impacts

# Climate and weather

- Climate: prevailing weather environmental conditions
  - Long term average
  - Relatively stable of weather
- Weather: conditions of atmosphere at particular time and space
  - Highly variable

# Weather is not climate



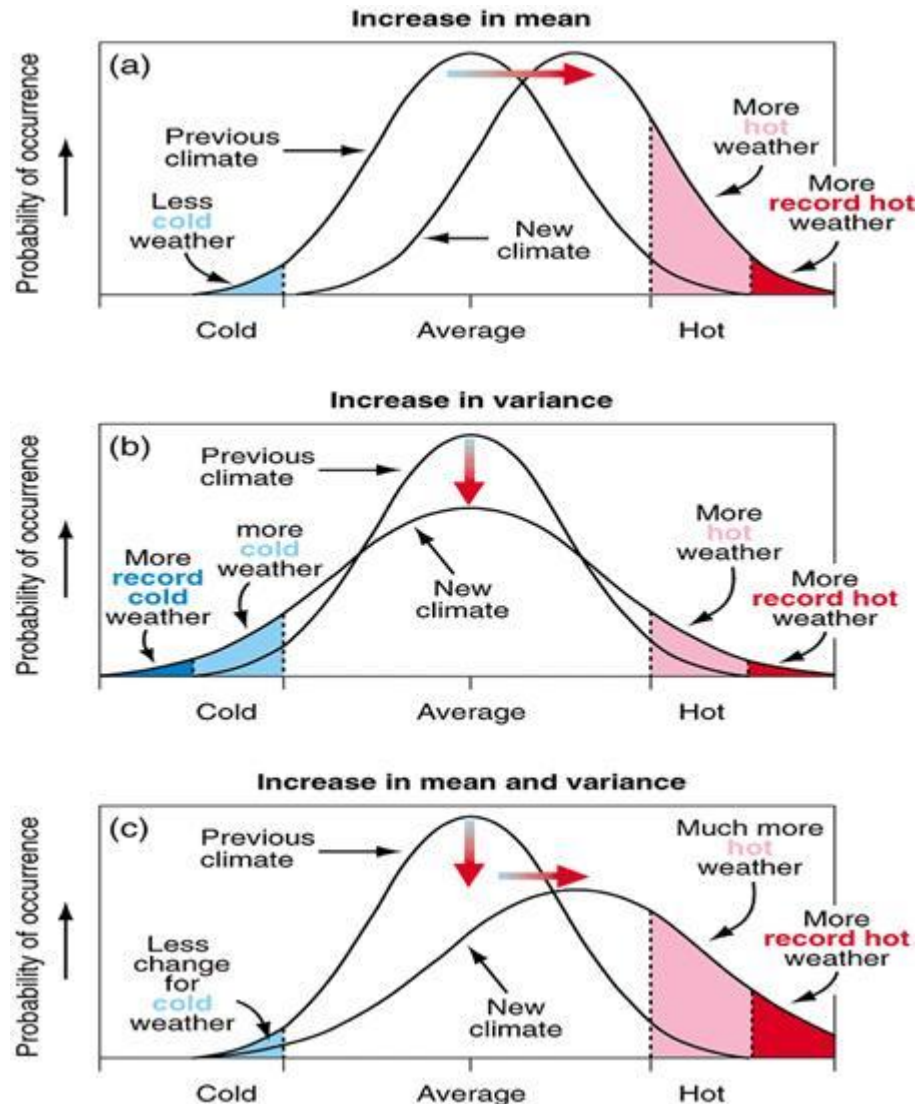
*"It's cold outside, so global warming can't be real."*



# Climate Variability and Climate Change

- Climate Variability: The range of fluctuations around long-term average climate conditions. It refers to variations in the state of climate beyond that of individual weather events.
  - Drivers of climate variability include El Nino Southern Oscillation (ENSO) and other phenomena. They produce warm, cool, wet, or dry periods across different regions and are a natural part of climate variability.
- Climate Change: A significant and persistent change in the mean state of the climate or its variability. Climate change occurs in response to changes in some aspect of Earth's environment.
  - It can include changes in temperature, precipitation, increase in occurrence of extreme weather events, as well as increased/decreased variability.

# Climate change impacts on weather



# Solar Emissive Power Flux

$E$  = emissive power flux ( $\text{W}/\text{m}^2$ )

$$A_{\text{sun}} = 4 \pi r_{\text{sun}}^2$$

$$A_{\text{orbit}} = 4 \pi R_{\text{orbit}}^2$$

$$T_{\text{sun}} = 5800 \text{ K}$$

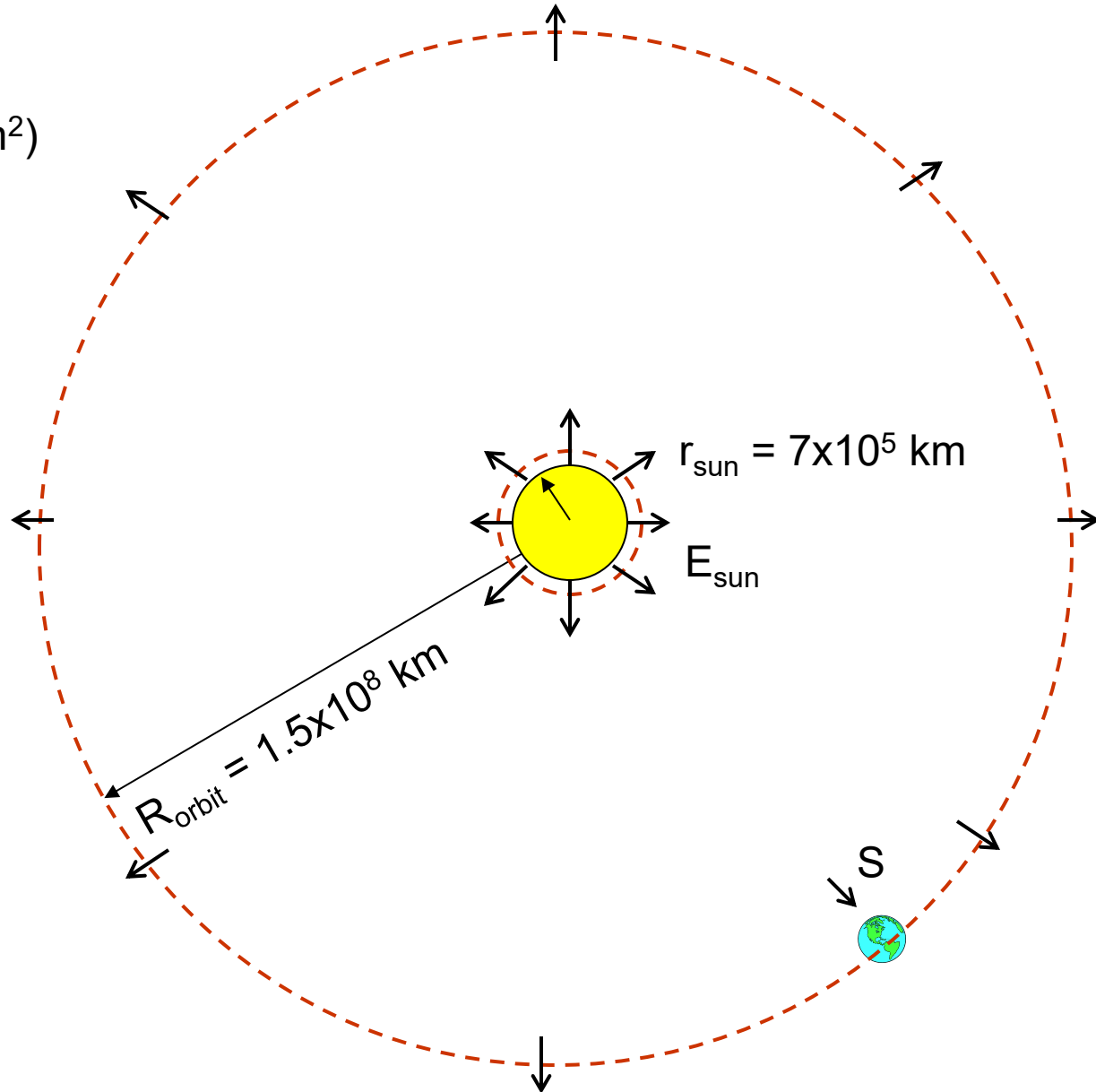
Try to estimate  
solar constant  $S$

ebe:

$$0 = A_{\text{sun}} E_{\text{sun}} - A_{\text{orbit}} S + 0 - 0$$

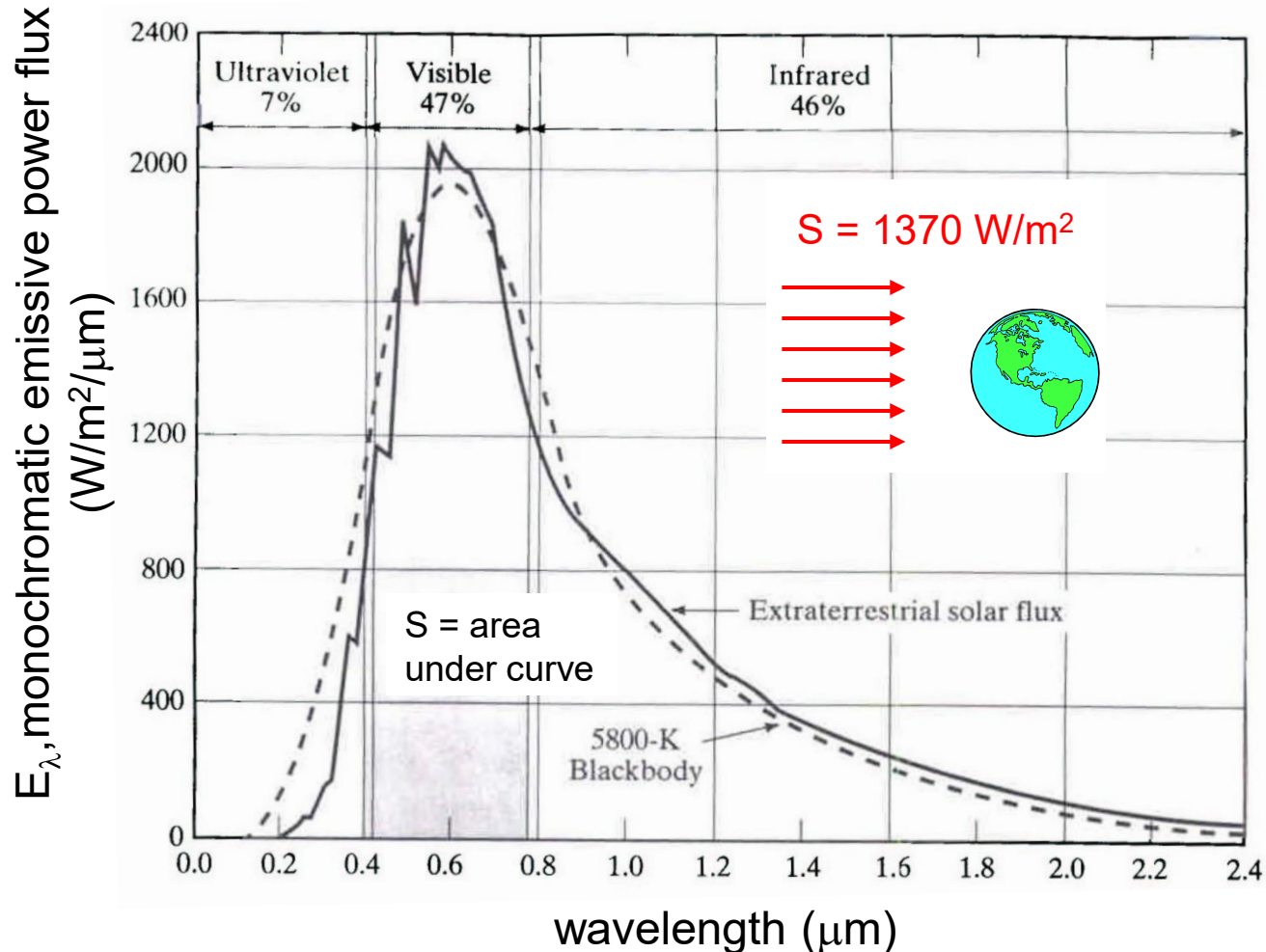
solution:

$$S = (r_{\text{sun}}/R_{\text{orbit}})^2 E_{\text{sun}}$$
$$= 2.15 \times 10^{-5} E_{\text{sun}}$$



# Solar Flux at Earth

From ebe:  $S = (r_{\text{sun}}/R_{\text{orbit}})^2 E_{\text{sun}} = \underline{\underline{1370 \text{ W/m}^2}}$

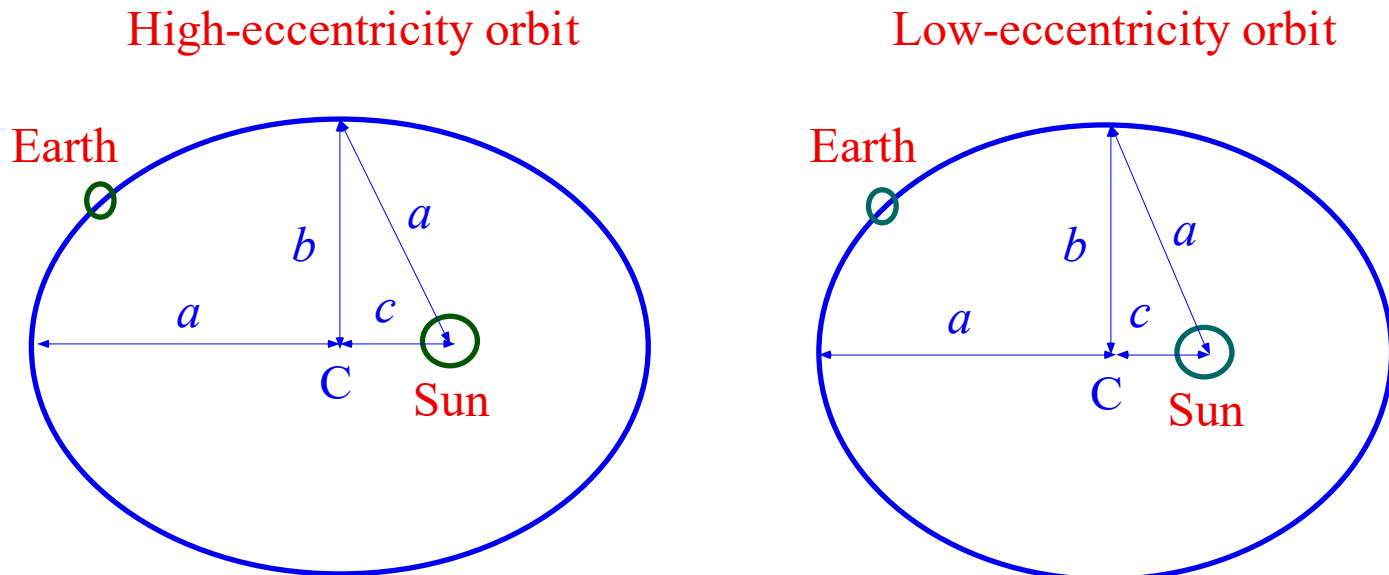


# Milankovitch Cycles

- Global wobbling is part of Milankovitch cycle
- The amount of solar radiation received by earth do vary substantially over long time period
- Earth's temperature varies naturally due to earth's movement relative to the sun
  - Eccentricity
  - Obliquity
  - Precession

# Eccentricity

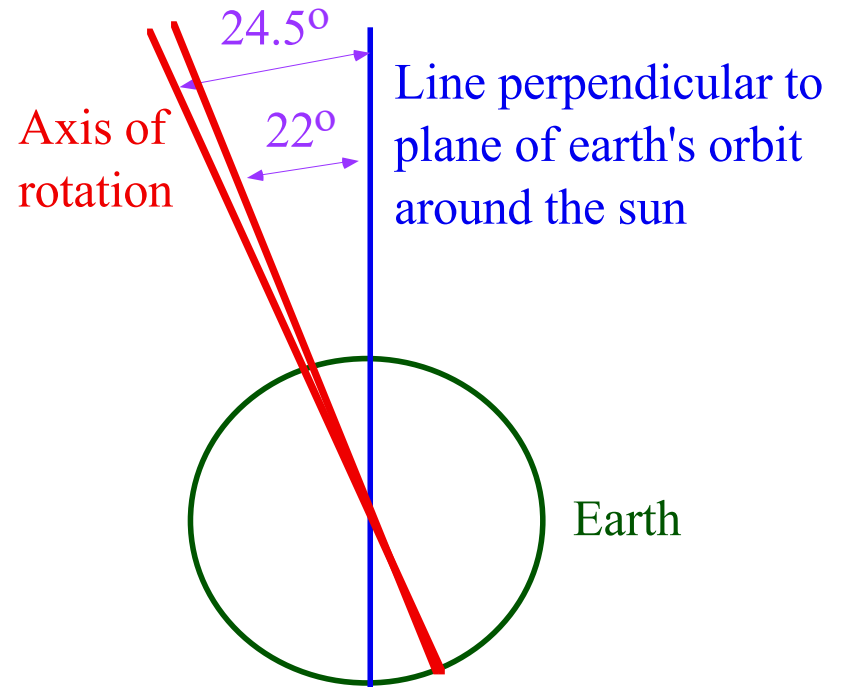
- Earth's orbit around the sun is elliptical
- The eccentricity ( $c/a$ ) varies from 0.01 – 0.05
  - Currently 0.017 and decreasing
- Roughly 100,000 year cycle



(Figure 12.21 Jacobson, 2012)

# Obliquity

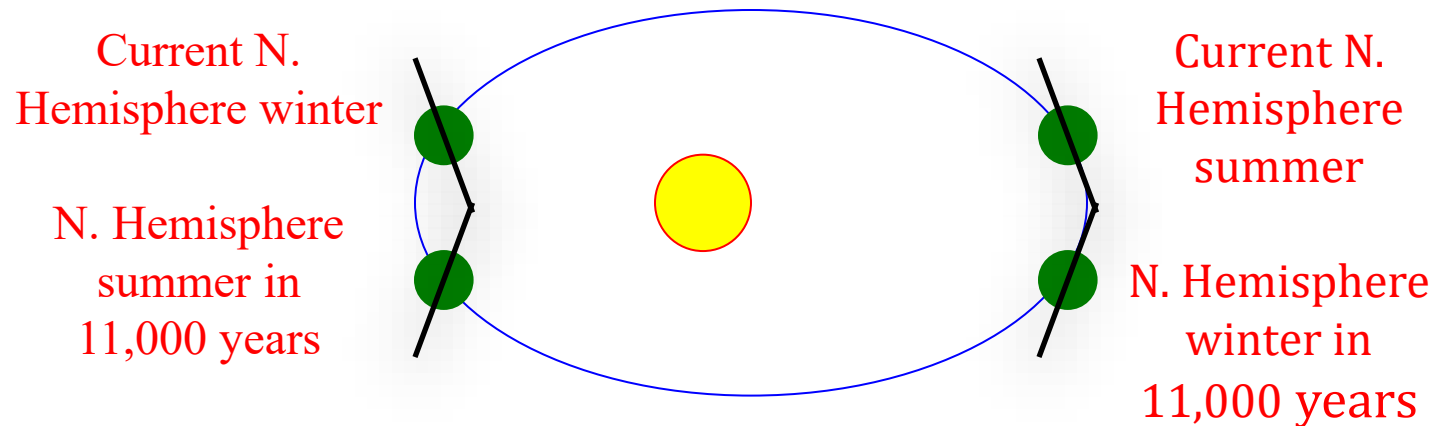
- The angle of the Earth's axial tilt
  - Current:  $23.5^\circ$  and decreasing
- 41,000 year cycle



(Figure 12.22 Jacobson, 2012)

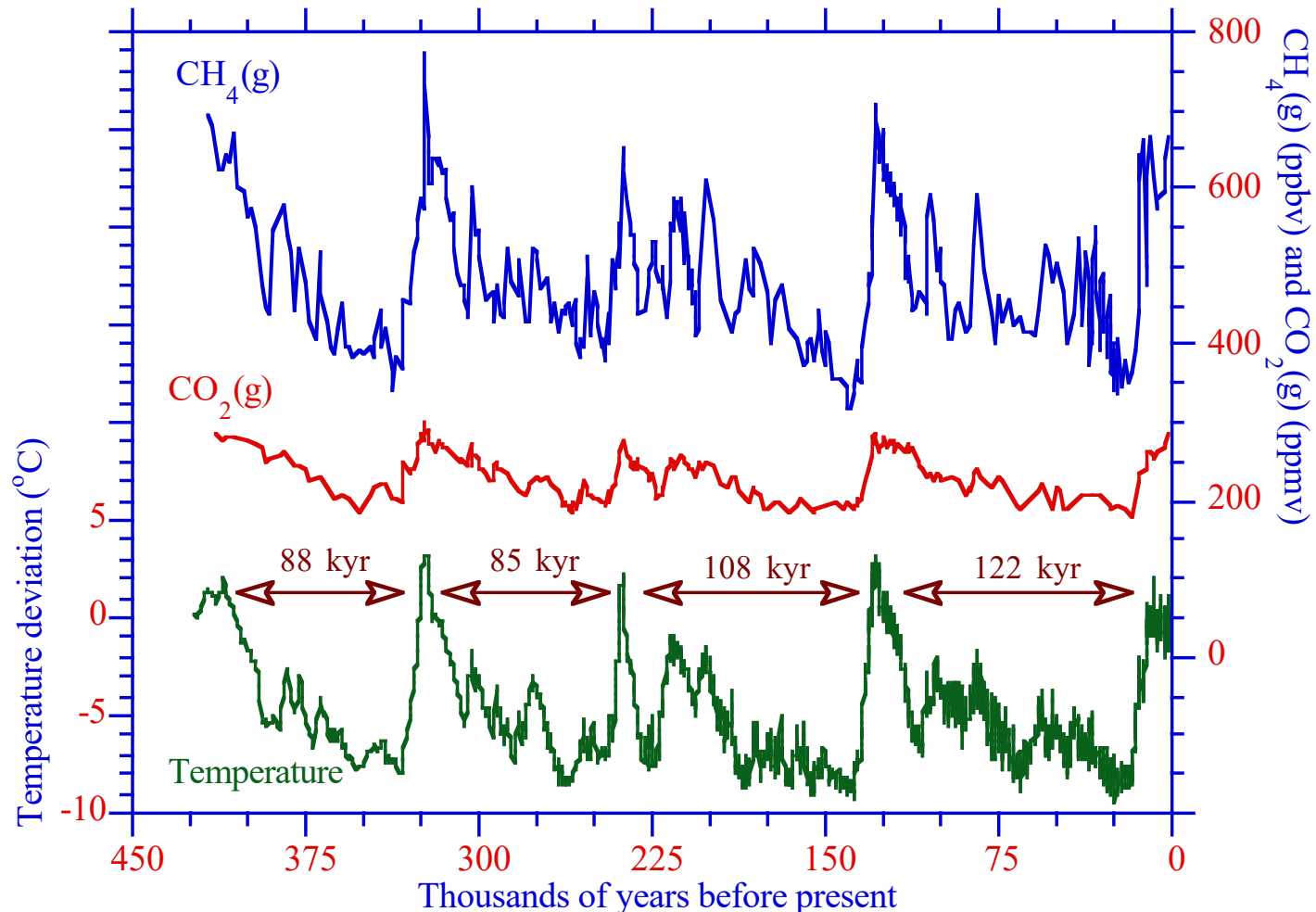
# Precession

- Earth's angular motion around axis
  - 22,000 years cycle



(Figure 12.22 Jacobson, 2012)

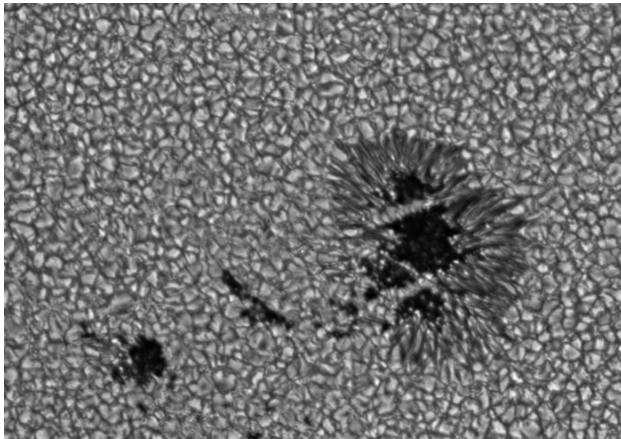
# Vostok ice core temperature data



100 thousand year time scale temp change (Figure 12.20 Jacobson, 2012)

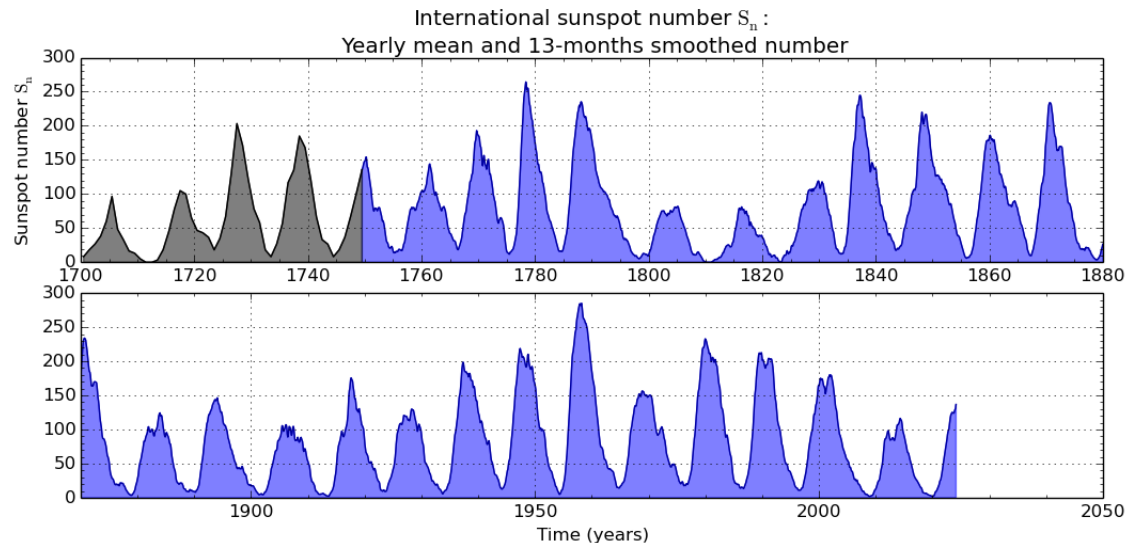
# Sunspot cycle

- Darker spot on the surface of the sun
- Affect solar radiation
- 11 year cycle



Sunspot

(<http://solarscience.msfc.nasa.gov/images/sunspot1.jpg>)



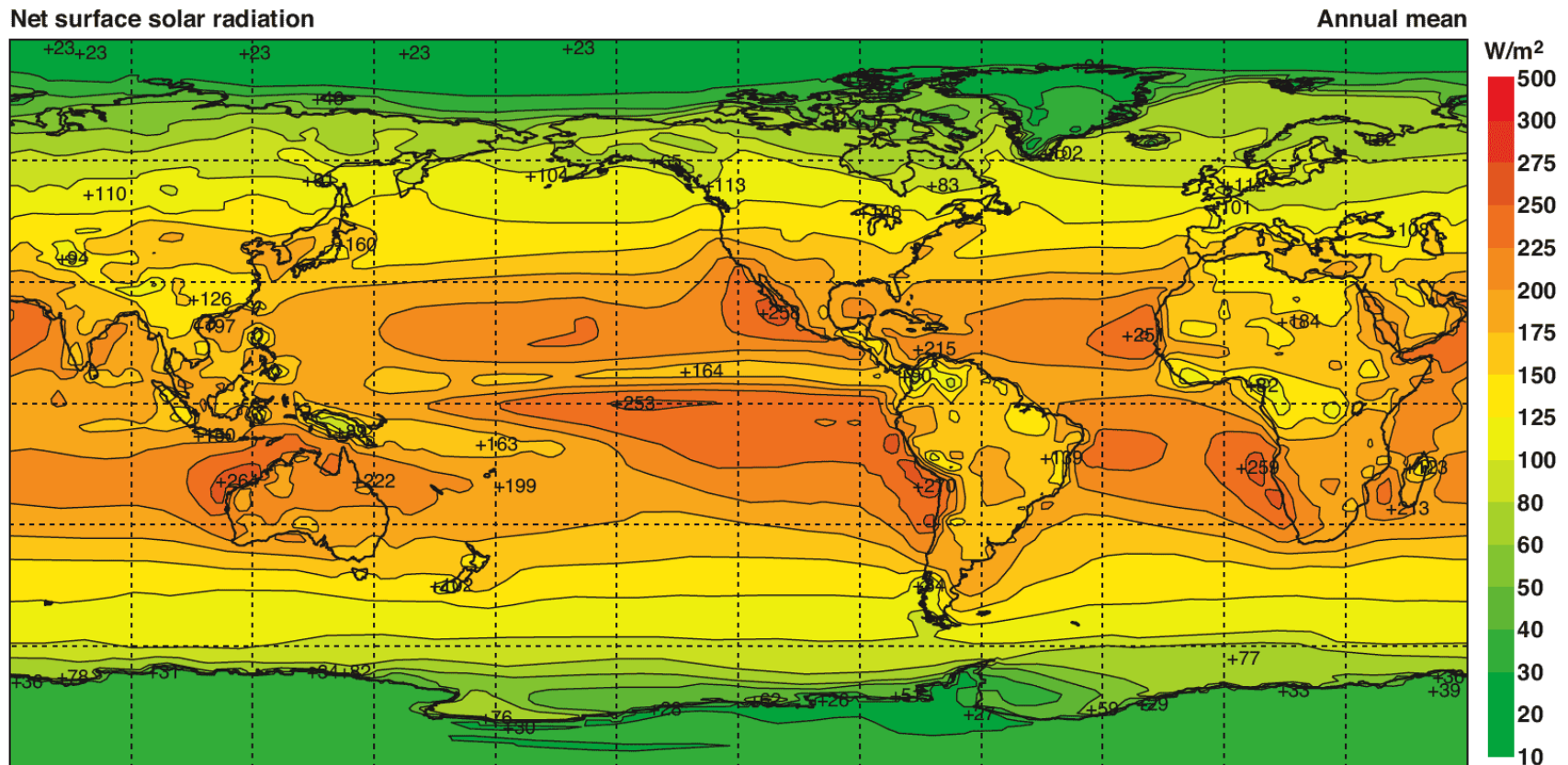
SILSO graphics (<http://sidc.be/silso>) Royal Observatory of Belgium 2024 September 1

Averaged sunspot count  
<https://www.sidc.be/SILSO/yearlyssnplot>

# Earth's energy balance

- Solar radiation provides energy to drive earth's climate system
  - UV, visible and IR
  - Energy flux at the top of the earth's atmosphere:
    - $\sim 1370 \text{ W/m}^2$ . The solar constant. Varies over time
    - Does not mean surface will radiation evenly
- Earth also emits radiation, at IR wavelength
  - Everything radiates, wavelength depends on temp

# Earth's energy balance



1979 – 2001 annual mean net surface fluxes of solar radiation  
(Kallberg et al 2005) ECMWF ERA-40 study

# Earth-Atmosphere-Sun System

avg solar flux over earth's surface  
=  $\sim 342 \text{ W/m}^2$

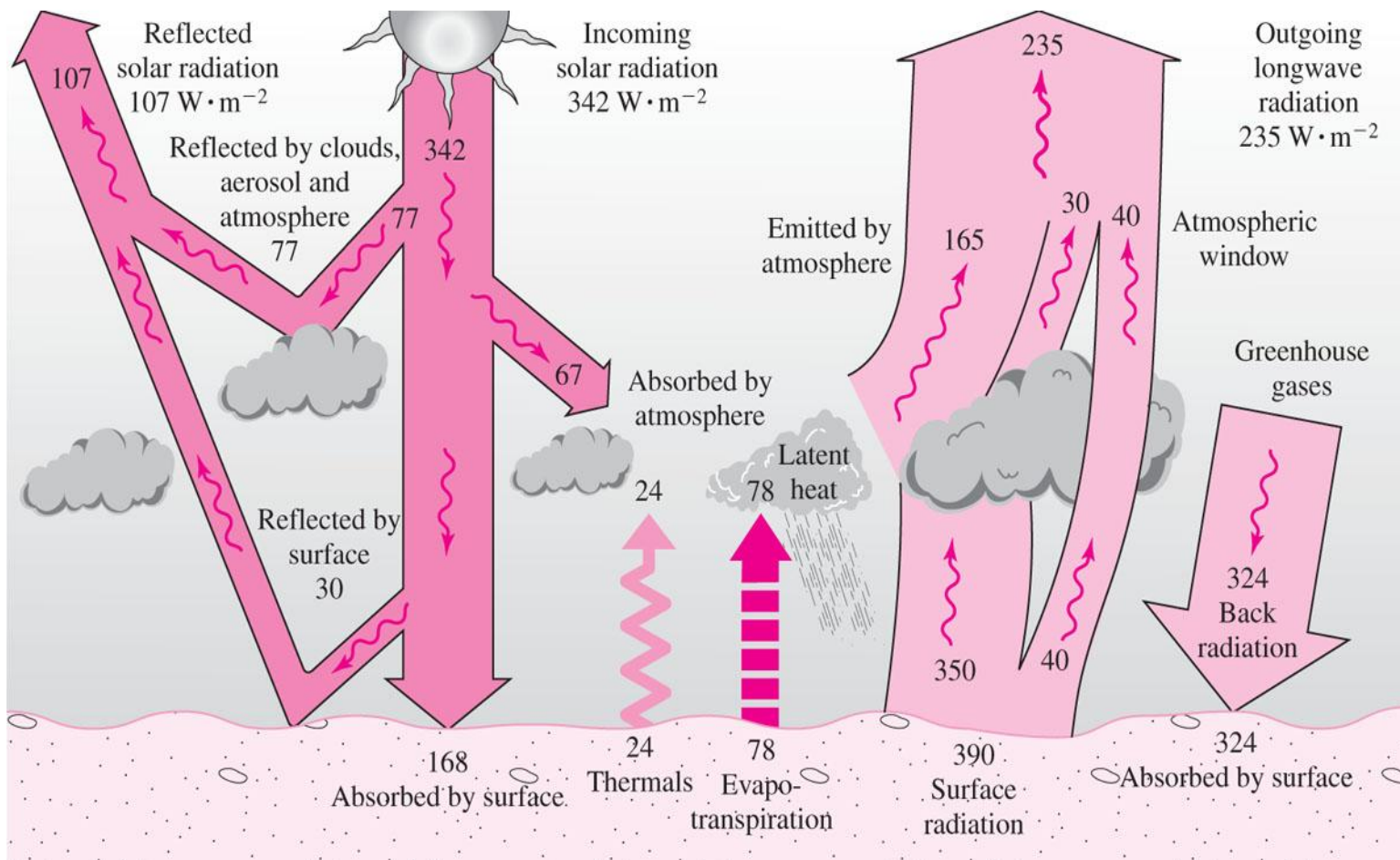
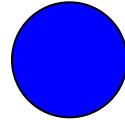


Fig 12-13

# Earth's energy balance

- Without the atmosphere, earth temperature would be below freezing
  - Assuming incoming radiation = outgoing IR radiation
- Actual average earth temperature is 15°C
- Differences due to natural greenhouse effect
  - Greenhouse gases absorb and re-emit IR radiation
  - Overall outgoing IR radiation < incoming radiation

# Earth Temperature



Venus

Earth

Mars

surface reflection

10%

10%

10%

$R_{\text{orbit}}$  (km)

$108 \times 10^6$

$150 \times 10^6$

$228 \times 10^6$

$S$  ( $\text{W}/\text{m}^2$ )

2620

1370

590

$T$  (w/o atm)

$46^\circ\text{C}$

$-2^\circ\text{C}$

$-53^\circ\text{C}$

relative size of atm

90

1

0.01

(molecules per surface area)

$T$  actual

$460^\circ\text{C}$

$15^\circ\text{C}$

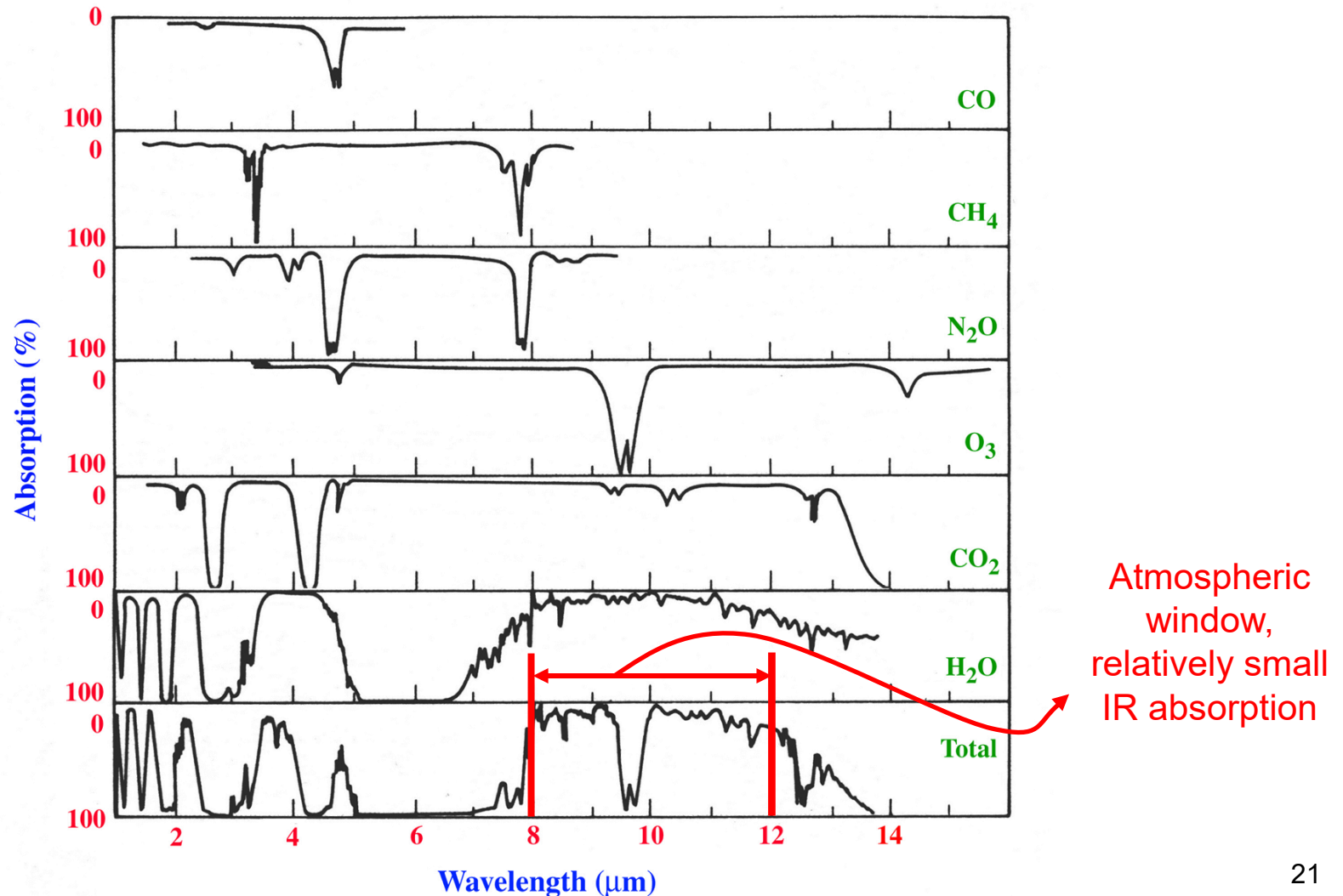
$-50^\circ\text{C}$

natural greenhouse effect

# Important compounds contribute to natural greenhouse effect

| Compound           | Contribution to <b>natural</b> greenhouse effect |
|--------------------|--------------------------------------------------|
| H <sub>2</sub> O   | 88.9%                                            |
| CO <sub>2</sub>    | 7.5%                                             |
| N <sub>2</sub> O   | 1.5%                                             |
| O <sub>3</sub>     | 1.1%                                             |
| CH <sub>4</sub>    | 0.5%                                             |
| CH <sub>3</sub> Cl | 0.3%                                             |
| Black carbon       | 0.2%                                             |

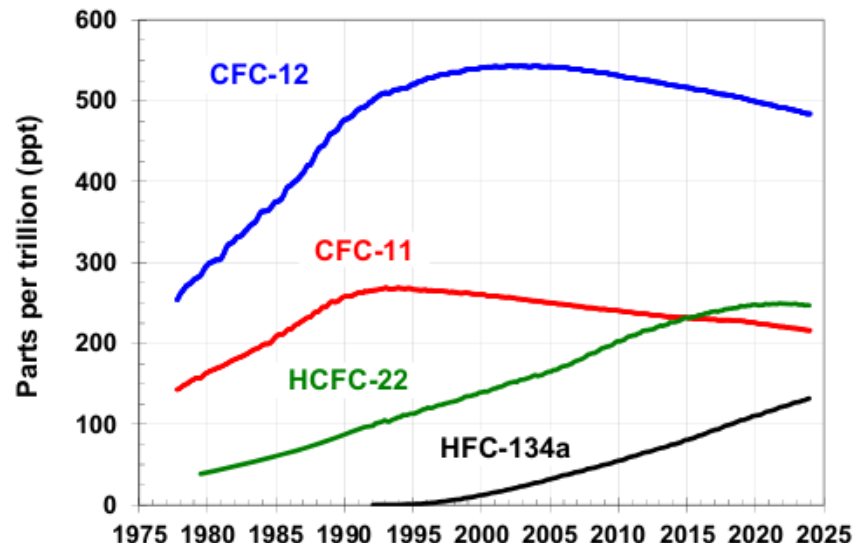
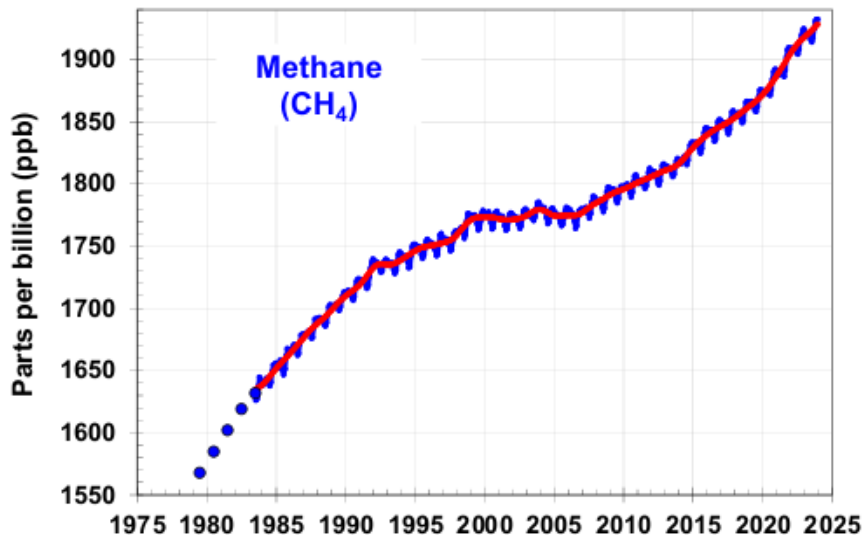
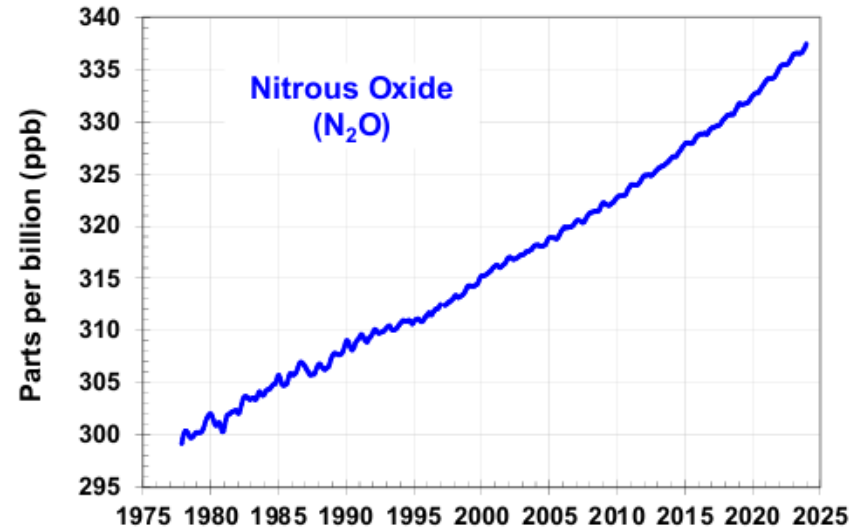
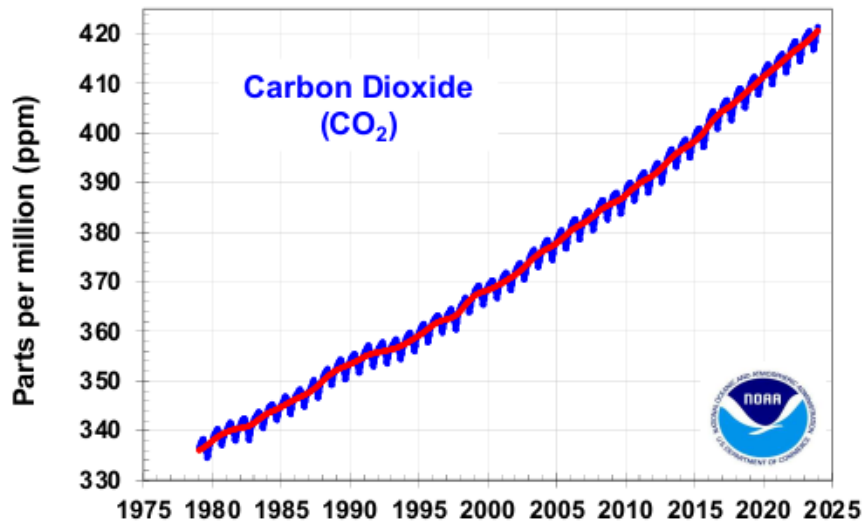
# Absorption of greenhouse gases



# Absorption of greenhouse gases

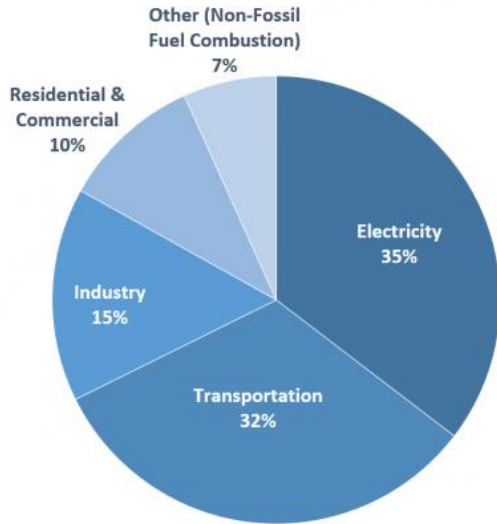
- Different gases absorb at different wavelength
- Water vapor: most important for natural greenhouse effect due to its abundance
- Overall, relatively small absorption at 8-12  $\mu\text{m}$ 
  - Atmospheric window: Atmosphere let these radiation escape away
- This is a simplified representation

# Anthropogenic Greenhouse Gas Trends

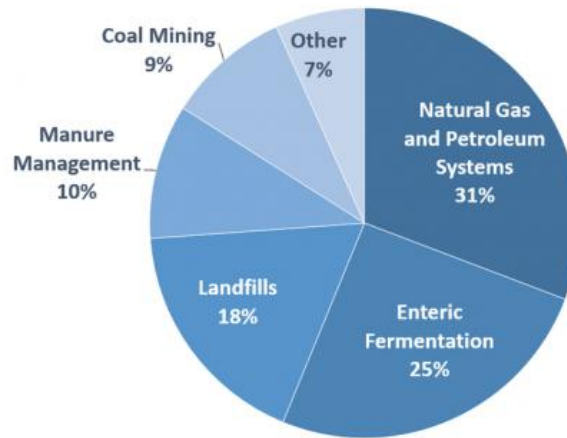


# US greenhouse emissions

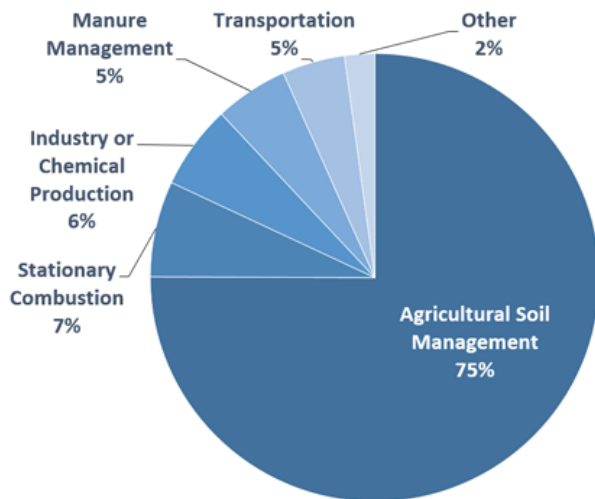
2015 U.S. Carbon Dioxide Emissions, By Source



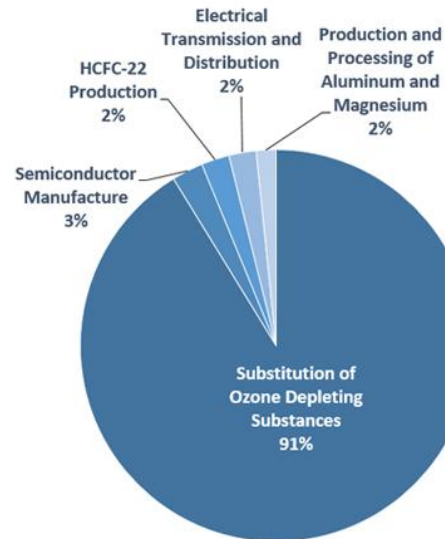
2015 U.S. Methane Emissions, By Source



2015 U.S. Nitrous Oxide Emissions, By Source



2015 U.S. Fluorinated Gas Emissions, By Source



# Climate system dynamics

- *Radiative forcing* can be used to quantify the strength of different changes on climate change
  - Change in net radiation  $\leftarrow$  introduction of an agent
  - Unit:  $\text{W/m}^2$
  - Net radiation: Incoming radiation – outgoing radiation
  - Typically evaluated at tropopause
- Solar radiation drives the climate system

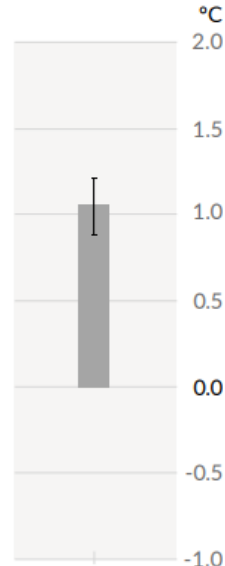
# Climate system dynamics

- A positive radiative forcing
  - Less outgoing IR radiation
  - Extra energy → earth warm up
  - E.g. greenhouse gases absorb outgoing IR radiation
- A negative radiative forcing
  - More outgoing IR radiation
  - Loss energy → earth cool down
  - E.g. some aerosols reflect sunlight

# Radiative forcing

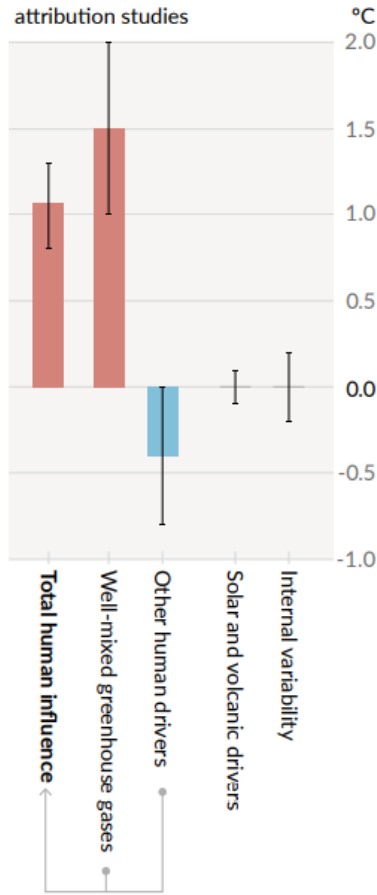
## Observed warming

a) Observed warming 2010-2019 relative to 1850-1900

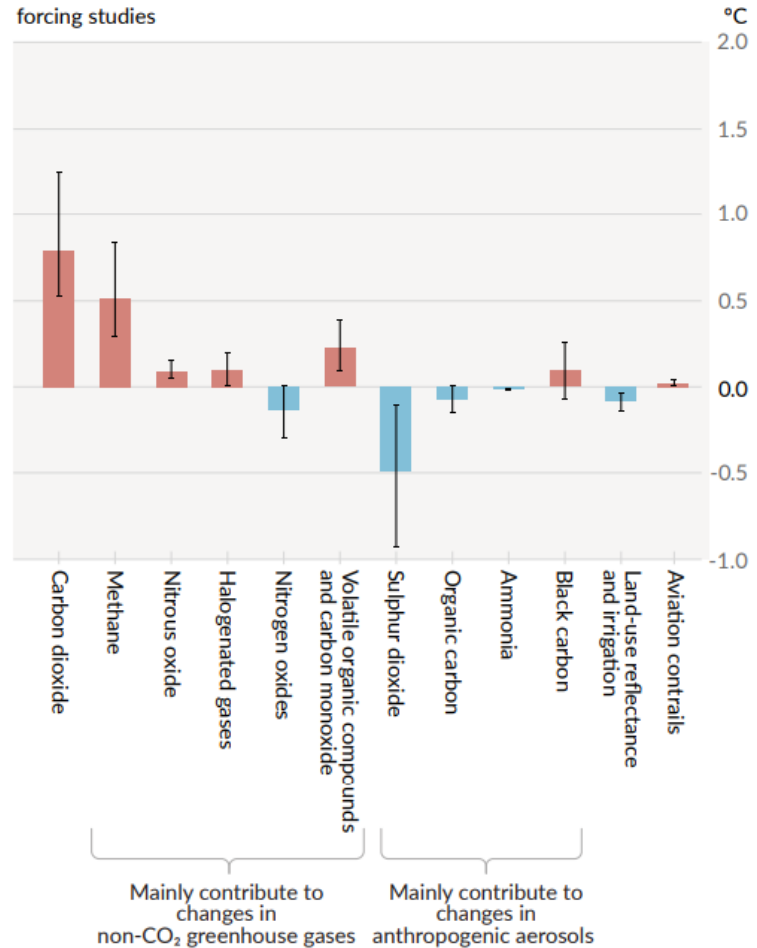


## Contributions to warming based on two complementary approaches

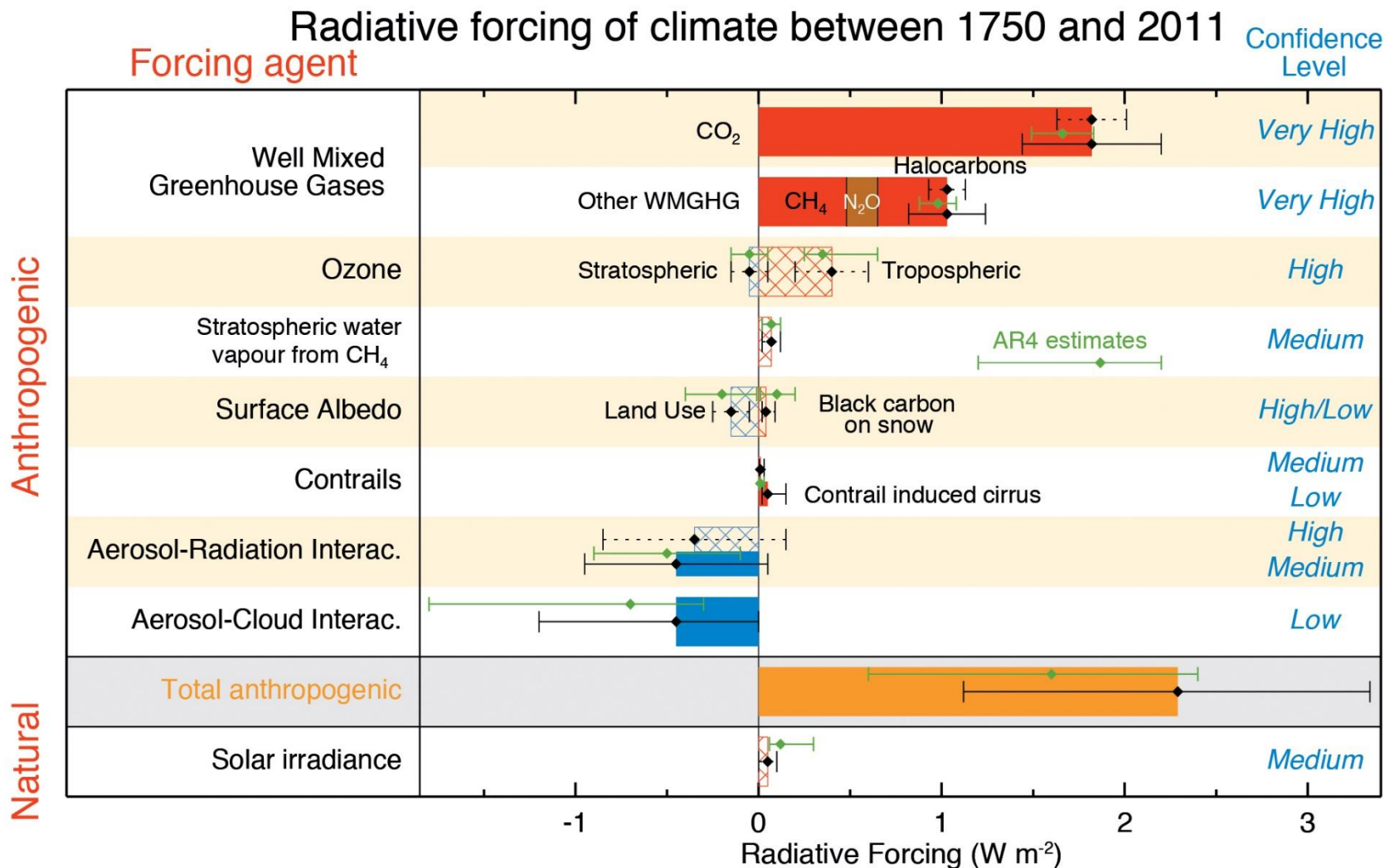
b) Aggregated contributions to 2010-2019 warming relative to 1850-1900, assessed from attribution studies



c) Contributions to 2010-2019 warming relative to 1850-1900, assessed from radiative forcing studies

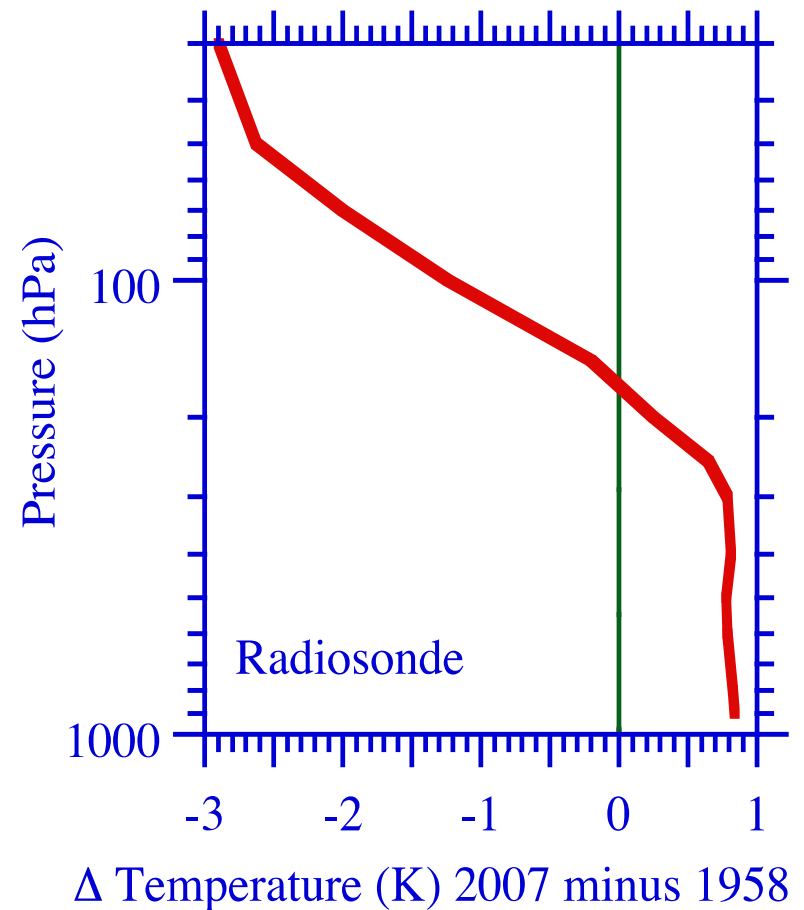


# Radiative forcing – AR5



# Vertical temperature structure

- Increased GHG at troposphere  
→ increased trapping of IR  
radiation from earth → less  
heat goes to stratosphere
- Ground Temp ↑
- Stratosphere Temp ↓



(Figure 12.14 Jacobson, 2012)

# Climate feedbacks

- Climate system responses to changes
- The results could increase/decrease radiative forcing
  - Positive feedback: enhance warming
  - Negative feedback: reduce warming

# Climate feedbacks

- Water vapor
  - P: Temp↑, evaporation↑, IR absorption↑, Temp↑
  - P: Temp↑, saturation vapor pressure↑, airborne water vapor↑, Temp↑
- Water vapor – cloud
  - P: Temp↑, more cloud ↑, IR absorption/radiation back↑, Temp↑
  - N: Temp↑, more cloud ↑, light reflection↑, Temp↓

# Climate feedbacks

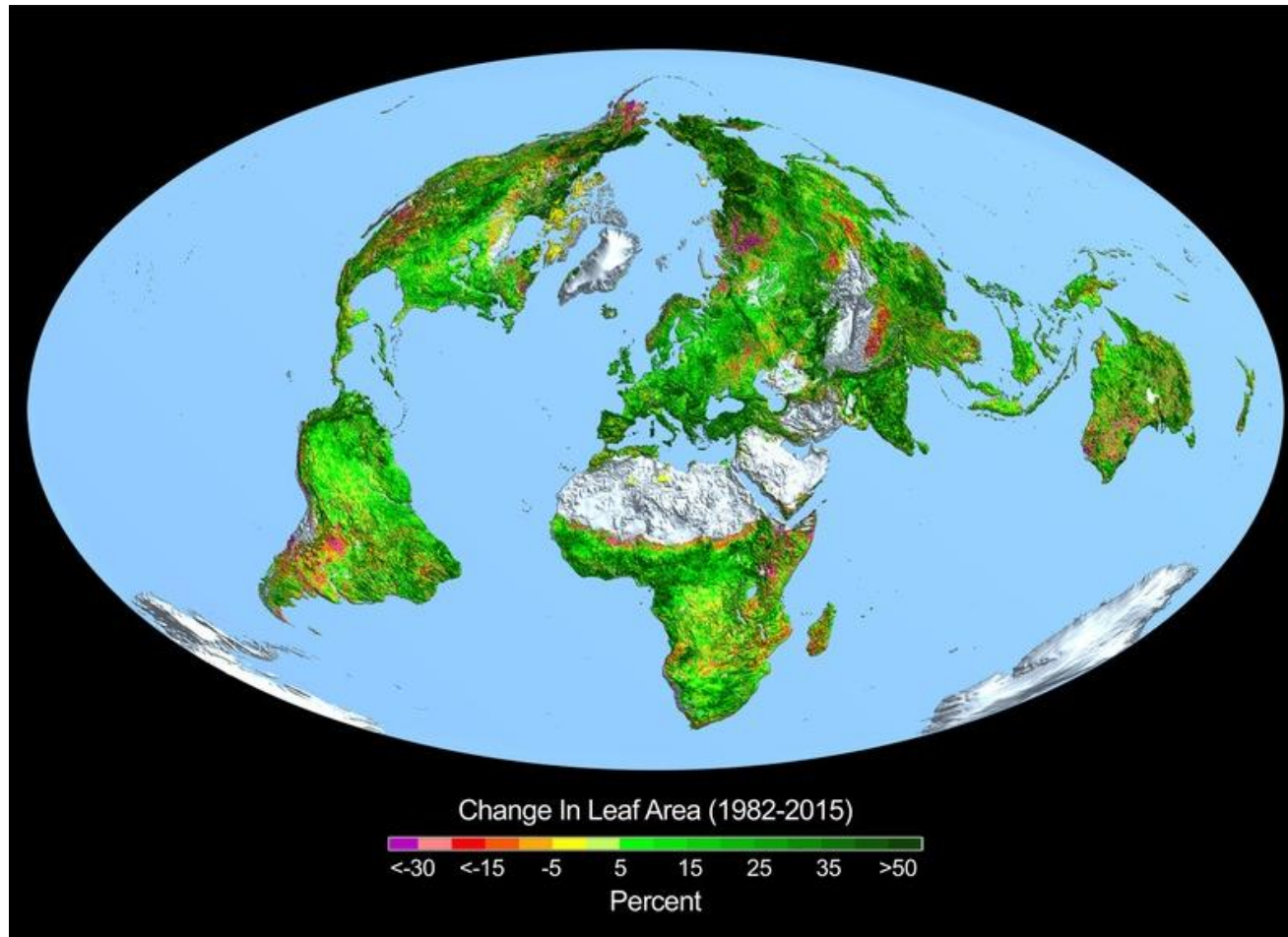
- Snow – albedo
  - Albedo: reflected radiation/incoming radiation
  - **P**: Temp↑, melt↑, exposed surface↑, reflection↓, solar radiation absorption↑, albedo↓, Temp↑
- Ocean – atmosphere
  - Ocean stores large amount of CO<sub>2</sub>
  - **P**: Temp↑, ocean CO<sub>2</sub> solubility↓, airborne CO<sub>2</sub>↑, Temp↑

# Climate feedbacks

- Bacteria – greenhouse gases
  - P: Temp↑, bacteria activity↑, GHG↑, Temp↑
- Plant – CO<sub>2</sub>
  - N: Temp↑, photosynthesis↑, CO<sub>2</sub>↓, Temp↓
- Permafrost – methane
  - P: Temp↑, Permafrost melt↑, CH<sub>4</sub>↑, Temp↑

# **Impact of Climate Change**

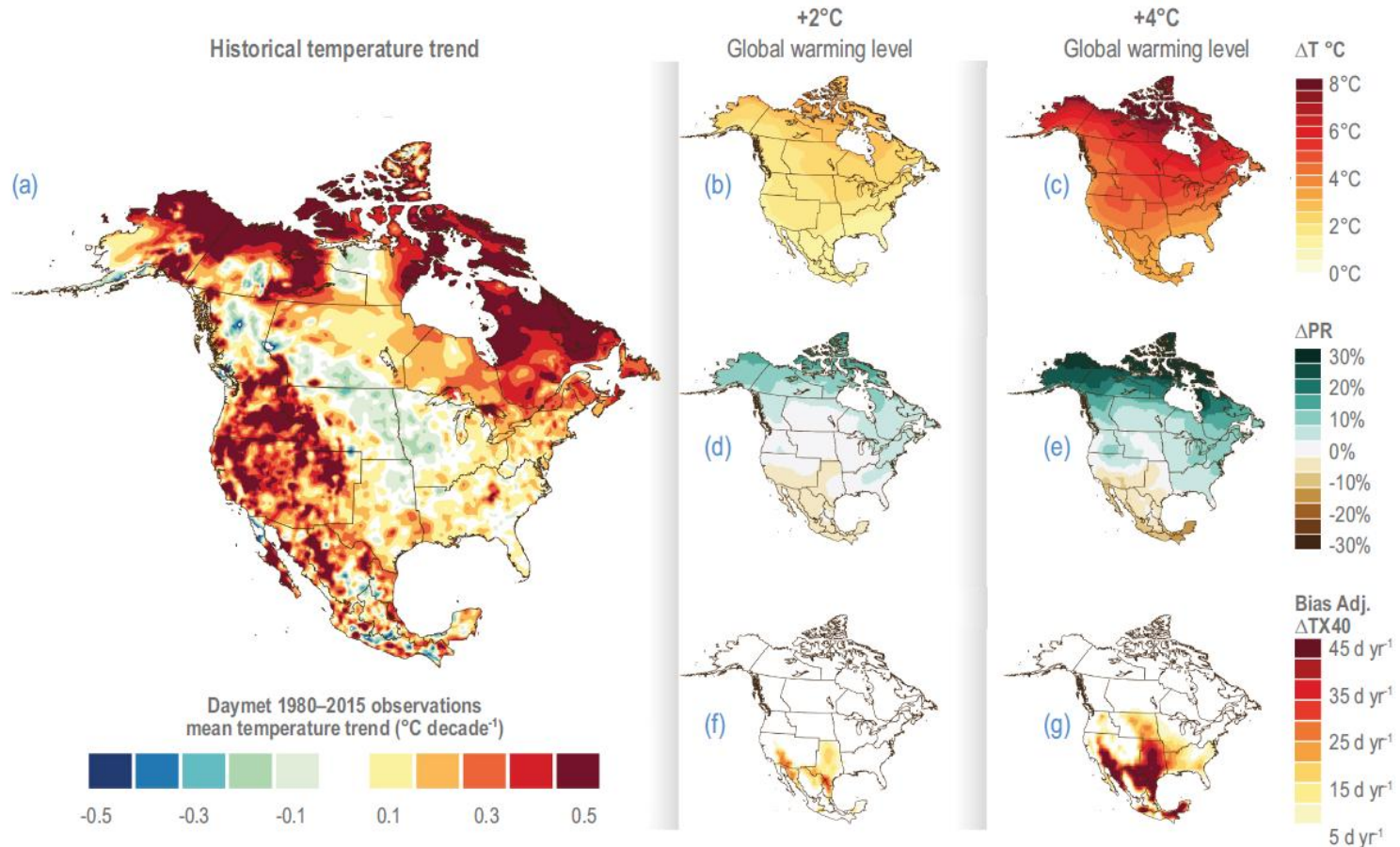
# Greening of earth



Source: <https://svs.gsfc.nasa.gov/12222>

# Extreme temperature

## Observed and projected climate changes across North America



**Figure 14.2 | Observed and projected climate changes across North America.** Black boundary lines delineate North American sub-regions (Figure 14.1). Data were extracted from Gutiérrez et al. 2021a via <http://interactive-atlas.ipcc.ch/> (WGI Interactive Atlas) (Gutiérrez et al., 2021b), where dataset details can be found. (A) Recent observations; (B) to (G) are from an ensemble of CMIP6 projections.

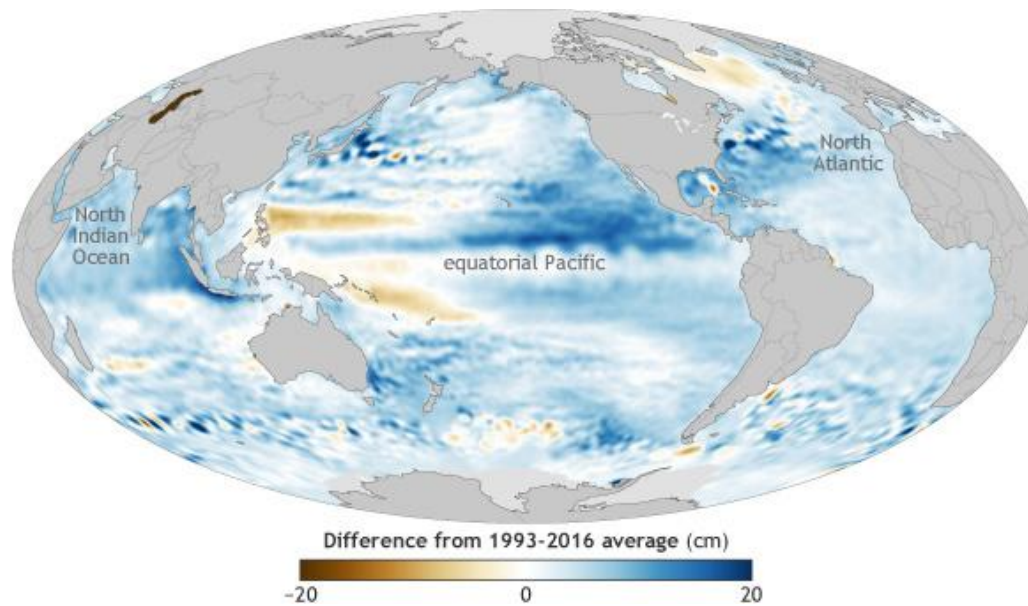
(A) Observed annual mean temperature trend over land for 1980–2015.

(B,C) Projected change in annual mean temperature over land relative to the 1986–2005 average, associated with 2°C or 4°C global warming.

(D,E) Like (B,C) but for projected percentage change in annual precipitation.

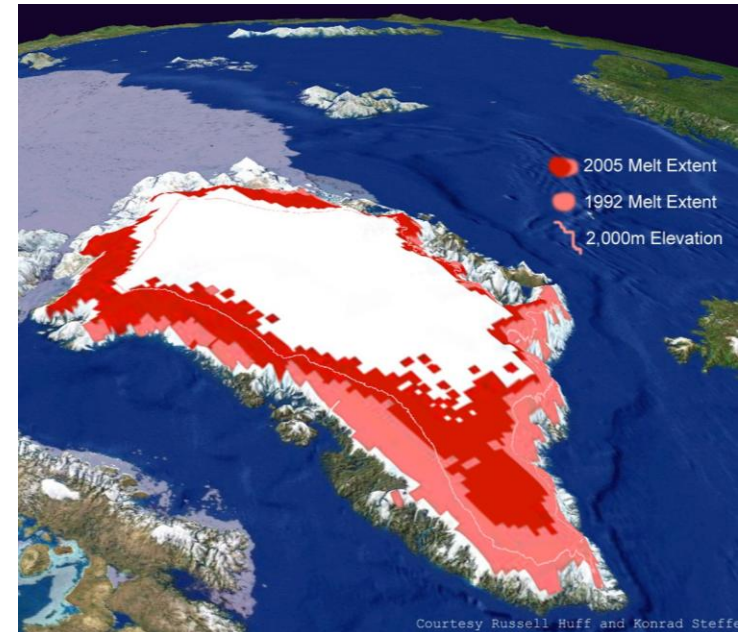
(F,G) Like (B,C) but for projected change in number of days per year with maximum temperature  $>40^{\circ}\text{C}$  ('TX40').

# Sea Level Rise



<https://www.climate.gov/news-features/understanding-climate/climate-change-global-sea-level>

# Greenland Ice Melt



<http://www.geo.arizona.edu/dgesl/index.html>

# Ocean pH changes

Source: [https://www.epa.gov/climate-indicators-climate-change-indicators-ocean-acidity](https://www.epa.gov/climate-indicators/climate-change-indicators-ocean-acidity)

